DataAnalysis

Gracie Lovell

2026-03-24

```r
library(tidyverse)
```

```
## Warning: package 'purrr' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(maps)
```

```
##
## Attaching package: 'maps'
##
## The following object is masked from 'package:purrr':
##
##     map
```

```r
library(randomForest)
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
##
##     margin
```

```r
merge <- read_csv("merged_dataset.csv")
```

```
## Rows: 98476 Columns: 16
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (3): price_display, cityname, state
## dbl (13): price, square_feet, latitude, longitude, avg_violent, avg_murder, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Linear Regression

Linear Model with all Crime Variables

```r
price_lm <- lm(price ~ square_feet + avg_violent + avg_murder + avg_rape + avg_robbery + avg_agrv_assaul
price_lm
```

```
##
## Call:
## lm(formula = price ~ square_feet + avg_violent + avg_murder +
##     avg_rape + avg_robbery + avg_agrv_assault + avg_prop + avg_burglary +
##     avg_larceny + avg_vehicle_theft, data = merge)
##
## Coefficients:
##       (Intercept)        square_feet        avg_violent         avg_murder
##         1379.9400             0.9615         -1529.3976          1468.6500
##          avg_rape        avg_robbery   avg_agrv_assault           avg_prop
##         1515.1929          1533.2098          1529.1847         -8603.6792
##      avg_burglary        avg_larceny  avg_vehicle_theft
##         8602.9775          8603.5187          8605.6014
```

```r
summary(price_lm)
```

```
##
## Call:
## lm(formula = price ~ square_feet + avg_violent + avg_murder +
##     avg_rape + avg_robbery + avg_agrv_assault + avg_prop + avg_burglary +
##     avg_larceny + avg_vehicle_theft, data = merge)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -34319   -342    -84    232  49793
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)         1.380e+03  1.524e+01   90.55   <2e-16 ***
## square_feet         9.615e-01  6.010e-03  159.99   <2e-16 ***
## avg_violent        -1.529e+03  1.236e+02  -12.37   <2e-16 ***
```

```
## avg_murder         1.469e+03  1.234e+02   11.90   <2e-16 ***
## avg_rape           1.515e+03  1.236e+02   12.26   <2e-16 ***
## avg_robbery        1.533e+03  1.237e+02   12.40   <2e-16 ***
## avg_agrv_assault   1.529e+03  1.236e+02   12.37   <2e-16 ***
## avg_prop          -8.604e+03  2.298e+02  -37.44   <2e-16 ***
## avg_burglary       8.603e+03  2.298e+02   37.44   <2e-16 ***
## avg_larceny        8.604e+03  2.298e+02   37.44   <2e-16 ***
## avg_vehicle_theft  8.606e+03  2.298e+02   37.45   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 728.7 on 98465 degrees of freedom
## Multiple R-squared:  0.3517, Adjusted R-squared:  0.3517
## F-statistic:  5342 on 10 and 98465 DF,  p-value: < 2.2e-16
```

Linear Model with Only Basic Crime Variables

```r
price_lm2 <- lm(price ~ square_feet + avg_violent + avg_prop, data = merge)
price_lm2
```

```
##
## Call:
## lm(formula = price ~ square_feet + avg_violent + avg_prop, data = merge)
##
## Coefficients:
## (Intercept)  square_feet  avg_violent     avg_prop
##   1809.3696       0.9668       0.2217      -0.3812
```

```r
summary(price_lm2)
```

```
##
## Call:
## lm(formula = price ~ square_feet + avg_violent + avg_prop, data = merge)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -34647   -388   -111    236  50300
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.809e+03  1.293e+01  139.94   <2e-16 ***
## square_feet  9.668e-01  6.402e-03  151.01   <2e-16 ***
## avg_violent  2.217e-01  1.949e-02   11.37   <2e-16 ***
## avg_prop    -3.812e-01  3.512e-03 -108.53   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 778.3 on 98472 degrees of freedom
## Multiple R-squared:  0.2604, Adjusted R-squared:  0.2604
## F-statistic: 1.156e+04 on 3 and 98472 DF,  p-value: < 2.2e-16
```

Linear Model with Only Specific Crime Variables

```r
price_lm3 <- lm(price ~ square_feet + avg_murder + avg_rape + avg_robbery + avg_agrv_assault + avg_burgl
price_lm3
```

```
##
## Call:
## lm(formula = price ~ square_feet + avg_murder + avg_rape + avg_robbery +
##     avg_agrv_assault + avg_burglary + avg_larceny + avg_vehicle_theft,
##     data = merge)
##
## Coefficients:
##       (Intercept)        square_feet         avg_murder           avg_rape
##         1473.9020             0.9578           -62.3411           -13.7338
##       avg_robbery   avg_agrv_assault       avg_burglary        avg_larceny
##            2.8800             0.1611            -0.5391            -0.2097
## avg_vehicle_theft
##            1.5922
```

```r
summary(price_lm3)
```

```
##
## Call:
## lm(formula = price ~ square_feet + avg_murder + avg_rape + avg_robbery +
##     avg_agrv_assault + avg_burglary + avg_larceny + avg_vehicle_theft,
##     data = merge)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -34278   -355    -92    241  49815
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)        1.474e+03  1.501e+01  98.226  < 2e-16 ***
## square_feet        9.578e-01  6.066e-03 157.896  < 2e-16 ***
## avg_murder        -6.234e+01  1.579e+00 -39.490  < 2e-16 ***
## avg_rape          -1.373e+01  2.277e-01 -60.326  < 2e-16 ***
## avg_robbery        2.880e+00  9.062e-02  31.781  < 2e-16 ***
## avg_agrv_assault   1.611e-01  3.482e-02   4.627 3.72e-06 ***
## avg_burglary      -5.391e-01  2.146e-02 -25.118  < 2e-16 ***
## avg_larceny       -2.097e-01  8.893e-03 -23.576  < 2e-16 ***
## avg_vehicle_theft  1.592e+00  2.561e-02  62.179  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 735.8 on 98467 degrees of freedom
## Multiple R-squared:  0.339,  Adjusted R-squared:  0.339
## F-statistic:  6313 on 8 and 98467 DF,  p-value: < 2.2e-16
```

Square footage is the strongest predictor of apartment price, with each additional square foot increasing rent by approximately $0.96. Certain violent crimes, such as murder and rape, show a strong, negative relationship

with rental prices, suggesting that more severe crime may reduce housing demand. Other offenses, such as burglary and larceny, have weaker, negative relationships, suggesting that they have a smaller impact on rent prices. However, other crimes such as robbery, aggravated assault, and vehicle theft show a positive relationship with price. This likely reflects underlying urban characteristics, where higher-cost cities tend to experience higher rates of certain crimes. Overall, the model suggests that both housing characteristics and crime patterns play a role in determining rental prices.

The detailed crime model ($R^2 = 0.339$) outperformed the aggregated crime model ($R^2 = 0.260$), indicating that disaggregated crime variables provide more explanatory power.

The model explains approximately 34% of the variation in apartment prices. While this may appear modest, it is consistent with expectations for real-world housing data, where prices are influenced by many unobserved factors such as location quality, amenities, and neighborhood characteristics. Therefore, the model provides meaningful insight despite not capturing all sources of variation.

Random Forests

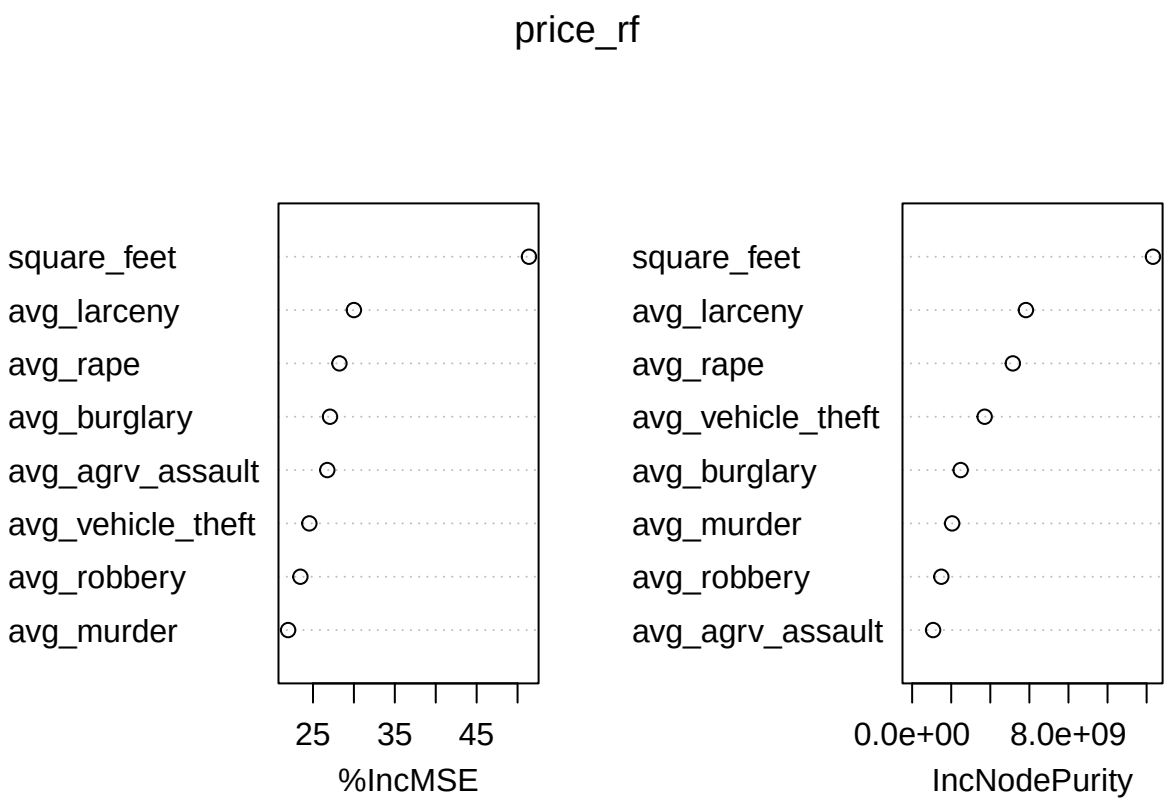
```
rf_data <- merge %>%
  select(price, square_feet, avg_murder, avg_rape, avg_robbery, avg_agrv_assault, avg_burglary, avg_larc
  na.omit()

price_rf <- randomForest(price ~ ., data = rf_data, ntree = 500, importance = TRUE)

print(price_rf)
```

```
##
## Call:
##  randomForest(formula = price ~ ., data = rf_data, ntree = 500,      importance = TRUE)
##                Type of random forest: regression
##                      Number of trees: 500
## No. of variables tried at each split: 2
##
##           Mean of squared residuals: 452472.6
##                     % Var explained: 44.76
```

```
importance(price_rf)
```

```
##                    %IncMSE IncNodePurity
## square_feet       51.39256   12324858255
## avg_murder        21.96275    2047230509
## avg_rape          28.23008    5153212030
## avg_robbery       23.45824    1493536145
## avg_agrv_assault  26.74690    1072624785
## avg_burglary      27.07902    2483656052
## avg_larceny       30.00110    5822769896
## avg_vehicle_theft 24.55589    3711565925
```

```
varImpPlot(price_rf)
```



The larger dataset is for predictive power and for maximizing explained variance. This model explains about 45% of the variation in apartment prices, which is 11% higher than our original linear model at 34%. The top four variables that improved accuracy (% Increase in MSE) were square_feet (50.39), avg_rape (30.00), avg_larceny (28.36), and avg_burglary (27.58). The top four variables that reduce prediction error (Increase in Node Purity) were square_feet (12,435,656,691), avg_rape (6,025,454,687), avg_larceny (5,028,460,816), and avg_vehicle_theft (3,442,330,822).

Obviously, square_feet is the most important predictor, which makes sense. Bigger apartments usually cost more.

Different types of crime affect rent differently, their impact on price does not follow a simple straight-line pattern.

K-Means Clustering

```
cluster_data <- merge %>%
  select(price, square_feet, avg_violent, avg_prop, latitude, longitude) %>%
  na.omit()

cluster_scaled <- scale(cluster_data %>% select(price, square_feet, avg_violent, avg_prop))
```

The smaller dataset is for simplicity, since we don't need specific crimes to see the high or low crime areas.

```r
set.seed(123)
price_km <- kmeans(cluster_scaled, centers = 3)
cluster_data$cluster <- price_km$cluster

aggregate(cluster_data, by = list(cluster = cluster_data$cluster), mean)
```

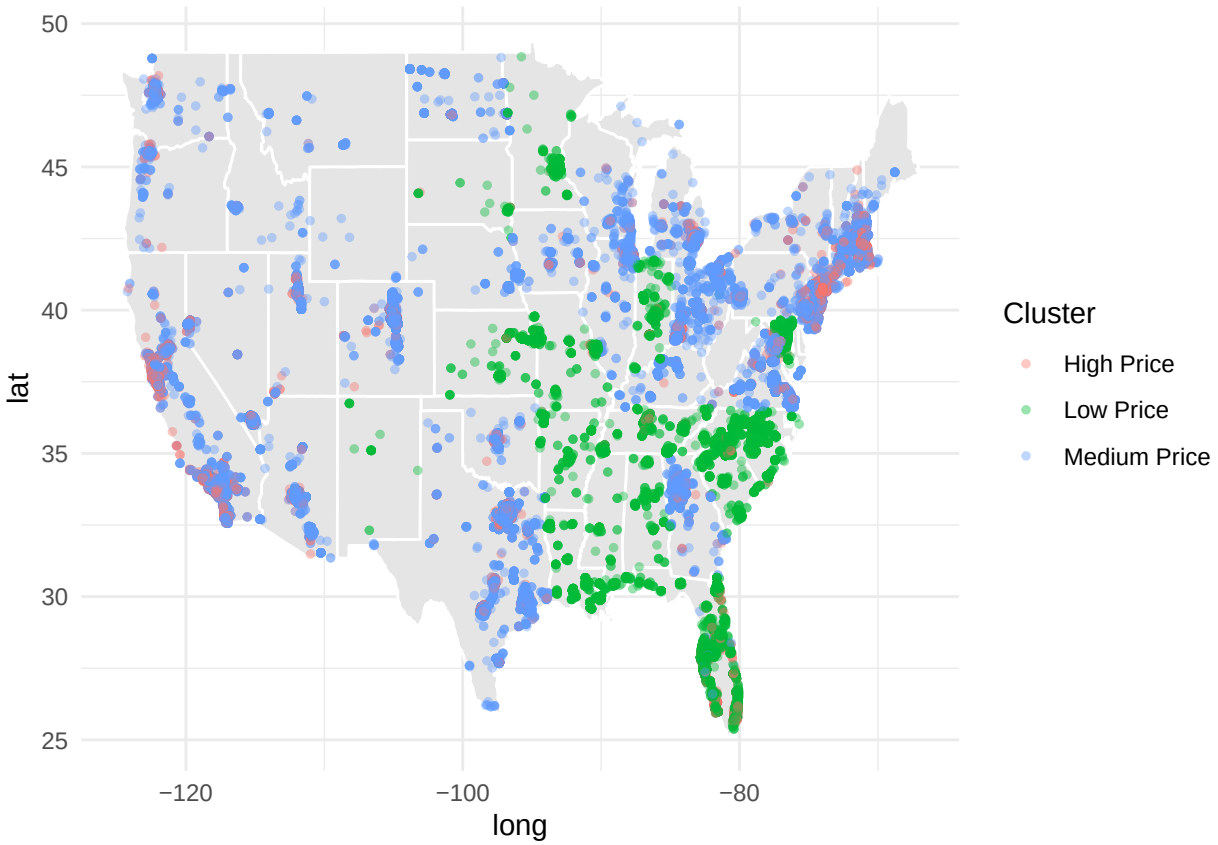
```
##   cluster    price square_feet avg_violent avg_prop latitude longitude cluster
## 1       1 2791.415   1353.9806    398.8938 2869.639 37.75194 -93.42242       1
## 2       2 1261.228    961.0773    648.9978 4239.209 35.02267 -82.92366       2
## 3       3 1310.071    851.8952    385.9141 3245.049 37.50703 -94.66602       3
```

Cluster 1 — high price, high sqft, medium violent crime, low property crime These are high-end apartments in safer areas with more space.

Cluster 2 — low price, medium sqft, high violent crime, high property crime These are cheaper apartments located in high-crime areas.

Cluster 3 — medium price, low sqft, low violent crime, medium property crime These are moderately priced apartments with mixed safety levels and smaller space.

```r
us_map <- map_data("state")

cluster_data <- cluster_data %>%
  filter(latitude > 25 & longitude > -125)

ggplot() +
  geom_polygon(data = us_map,
               aes(x = long, y = lat, group = group),
               fill = "gray90", color = "white") +
  geom_point(data = cluster_data,
             aes(x = longitude, y = latitude, color = factor(cluster, labels = c("High Price", "Low Pri
             alpha = 0.4, size = 1) +
  labs(color = "Cluster") +
  theme_minimal()
```


50
45
40
35
30
25
lat
−120
−100
−80
long
Cluster
High Price
Low Price
Medium Price